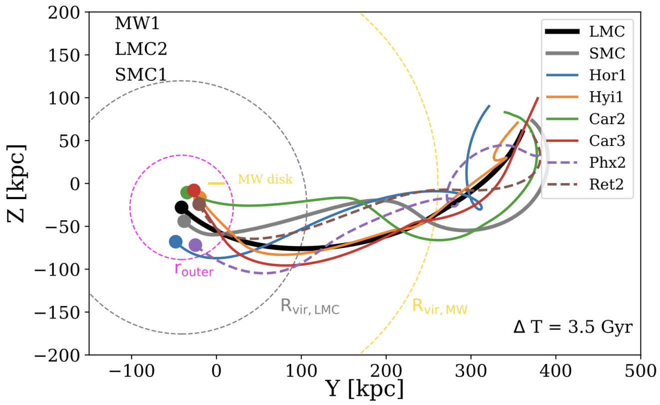

Figure 3: Orbits of all satellites identified as short-term (Ret2, Phx2; dashed lines) and long-term Magellanic satellites (Car2, Car3, Hor1, Hyi1; solid lines) using the fiducial LMC model for the last 3.5 Gyr projected in the YZ-galactocentric plane [35]. The disk of the MW lies along the z-axis. Filled circles represent the positions of all satellites today. All satellites evidence a shared orbit with the LMC (black line), providing the first evidence for a true "satellites of satellites" hierarchy.

to use the 6D phase space properties of M31's most massive satellite, M33, to estimate a cosmologically motivated M31 mass of $1.37^{+1.39}_{-0.75} \times 10^{12}\, M_\odot$. Since then, 6D phase space information has become available for additional M31 satellites, and in Patel 2022 (in prep.), I find a preliminary mass of $2.81^{+1.48}_{-0.75} \times 10^{12}\, M_\odot$ using four satellites, reducing the uncertainty of using just one satellite galaxy by more than half. This high mass has direct implications for understanding the nature of M31's satellite system as a population, its accretion history, and for the timing of major LG events both past and future to name a few (see Proposed Research).

1.4 Identifying Satellites of the Magellanic Clouds (Patel et al. 2020 – P20)

Many recently discovered dwarf satellite galaxies around the MW are located in close proximity to the MCs [e.g., 1, 8, 25, 24, 23, 9, 45, 46, 47]. Using *Gaia* DR2 PMs, I determined which satellite galaxies are most likely to have a Magellanic origin by rewinding the orbital histories of all galaxies in a combined MW+LMC+SMC potential, including dynamical friction from both the MW and LMC [P20 35]. I calculated thousands of orbits iterating over measurement uncertainties in the satellite phase space properties and developed a new set of metrics to identify "satellites of a satellite" using the average distance and velocity at pericenter relative to the properties of the LMC. I concluded that approximately six satellites were accreted into the MW's halo with the MCs, as illustrated in Figure 3. This is the first confirmation of a satellites of satellites hierarchy and is complementary to ongoing projects centered on M33 satellites (see §1.2).

Shortly thereafter, Sacchi et al. [38] used my classifications of LMC satellites to study the star formation histories (SFHs) of seven ultra-faint dwarfs, concluding that LMC satellites tend to show prolonged star formation (SF) and thus a later quenching time compared to non-LMC satellites, implying a connection between quenching time and environment. I am currently collaborating with N. Garavito-Camargo (Flatiron/CCA) to combine my orbital modeling techniques with his basis field expansion characterization of the MW-LMC system [15] to calculate the most accurate orbits of all MW satellites to date, including the effects of the LMC's dark matter wake, its tidal debris, and the large-scale under- and overdensities it imparts on the MW's halo (Fig. 5; Patel et al. 2023, in prep.).

My past work uses the combined power of high precision astrometric data sets with high-resolution simulations of the Universe to (1) understand the influence of massive satellite galaxies on their hosts and their low mass companions; (2) look for instances of small-scale clustering via orbit reconstruction; (3) deepen our understanding of the "satellites of satellites" hierarchy; and (4) to use satellite galaxies as a tool for constraining properties of their hosts. Together these projects have revealed a modified view of the LG and its history, overturning conventional wisdom about